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Ocenjevanje volatilnosti z Markovskim prehajanjem na finančnih trgih s sekvenčnimi Monte Carlo metodami

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Estimating Markov switching volatility in financial markets using sequential Monte Carlo methods

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Worth mentioning in the acknowledgment is everyone who contributed to your thesis.

Vito Levstik, 2026

To all the flowers of this world.

"The only reason for time is so that everything doesn't happen at once."

— Albert Einstein

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List of used acronyms

acronym	meaning
CA	classification accuracy
DBMS	database management system
SVM	support vector machine
...	...

Abstract

Title: Estimating Markov switching volatility in financial markets using sequential Monte Carlo methods

This sample document presents an approach to typesetting your BSc thesis using LaTeX. A proper abstract should contain around 100 words which makes this one way too short. A good abstract contains: (1) a short description of the tackled problem, (2) a short description of your approach to solving the problem, and (3) (the most successful) result or contribution in your thesis.

Keywords

computer, computer, computer

Povzetek

Naslov: Ocenjevanje volatilnosti z Markovskim prehajanjem na finančnih trgih s sekvenčnimi Monte Carlo metodami

V vzorcu je predstavljen postopek priprave magistrskega dela z uporabo okolja LaTeX. Vaš povzetek mora sicer vsebovati približno 100 besed, ta tukaj je odločno prekratek. Dober povzetek vključuje: (1) kratek opis obravnavanega problema, (2) kratek opis vašega pristopa za reševanje tega problema in (3) (najbolj uspešen) rezultat ali prispevek magistrske naloge.

Ključne besede

računalnik, računalnik, računalnik

Chapter 1

Introduction

Chapter 2

Stochastic Volatility

Volatility is a fundamental concept in financial econometrics and is commonly used as a measure of the variability or uncertainty of asset returns. In its most basic form, volatility refers to the standard deviation of returns over a given time horizon, and therefore quantifies the magnitude of fluctuations around the expected return.

Many models assume that volatility is constant. In practice, however, financial markets exhibit periods of high volatility, for example, during financial crises, and periods of low volatility, usually during stable economic conditions. Stochastic volatility models capture this behavior by allowing volatility to vary over time according to a stochastic process.

This chapter begins by introducing volatility and then develops the state-space formulation used throughout the thesis, with particular emphasis on the Markov switching stochastic volatility (MSSV) model.

2.1 Volatility

Let P_t denote the price of an asset (stock, index, commodity, etc.) at time t. The log return of the asset between two consecutive time points is defined as

$$Y_t = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right). \tag{2.1.1}$$

The advantage of using log returns rather than simple returns lies in their additive structure over time and their natural treatment of compounding effects [1]. Thus, throughout this thesis, the term return refers to a log return.

While returns are directly observable, volatility is a characteristic of the underlying return distribution and is not directly observable from a single realization. At each time point t, only one return is observed, whereas volatility is defined through the variability of the entire return-generating distribution. Consequently, time-varying volatility must be inferred from data rather than measured directly.

To formalize this idea, we define the conditional mean and variance of the return as

$$\mu_t = E[Y_t \mid \mathcal{F}_{t-1}], \tag{2.1.2}$$

$$\sigma_t^2 = \mathrm{Var}(Y_t \mid \mathcal{F}_{t-1}) = E[(Y_t - \mu_t)^2 \mid \mathcal{F}_{t-1}], \tag{2.1.3}$$

where $\mathcal{F}_{t-1}$ denotes the information set containing all relevant information available up to time $t-1$ [2]. The quantity σ_t is referred to as the conditional volatility. In many financial applications, the conditional mean is assumed to be negligible relative to the variability of returns, so that the focus is primarily on modelling the conditional volatility process.

So far, we have not specified the time scale of the returns. For example, Y_t could represent daily, weekly, or monthly returns, and t could therefore index days, weeks, or months. In this thesis, we focus on daily returns, specifically close-to-close daily returns, meaning that P_t is the closing price of the asset on day t[1]. The goal is then to estimate the daily conditional volatilities σ_t.

When high-frequency intraday data are available, the latent conditional variance can be approximated using realized variance. Consider a day t, divided into m equally spaced intraday intervals. Let $Y_{t,i}^{(m)}$ denote the return over the i-th intraday interval, $i = 1, \ldots, m$, so that the corresponding daily

[1] One could also consider open-to-open returns, where P_t is the opening price of the asset on day t.

return can be written as

$$Y_t = \sum_{i=1}^{m} Y_{t,i}^{(m)}. \tag{2.1.4}$$

It is often reasonable to assume that

$$E\left[Y_{t,i}^{(m)} \mid \mathcal{F}_{t-1}\right] \approx 0, \tag{2.1.5}$$

$$\mathrm{Cov}\left(Y_{t,i}^{(m)}, Y_{t,j}^{(m)} \mid \mathcal{F}_{t-1}\right) \approx 0, \qquad \text{for } i \neq j. \tag{2.1.6}$$

Under these assumptions, we have

$$\sigma_t^2 = \mathrm{Var}(Y_t \mid \mathcal{F}_{t-1}) = \mathrm{Var}\left(\sum_{i=1}^{m} Y_{t,i}^{(m)} \,\middle|\, \mathcal{F}_{t-1}\right) \tag{2.1.7}$$

$$\approx \sum_{i=1}^{m} \mathrm{Var}\left(Y_{t,i}^{(m)} \mid \mathcal{F}_{t-1}\right) \tag{2.1.8}$$

$$\approx \sum_{i=1}^{m} E\left[\left(Y_{t,i}^{(m)}\right)^2 \,\middle|\, \mathcal{F}_{t-1}\right] \tag{2.1.9}$$

$$= E\left[\sum_{i=1}^{m} \left(Y_{t,i}^{(m)}\right)^2 \,\middle|\, \mathcal{F}_{t-1}\right] \tag{2.1.10}$$

$$= E\left[RV_t^{(m)} \mid \mathcal{F}_{t-1}\right], \tag{2.1.11}$$

where we defined the realized variance at sampling frequency m as

$$RV_t^{(m)} = \sum_{i=1}^{m} \left(Y_{t,i}^{(m)}\right)^2. \tag{2.1.12}$$

Thus, under these assumptions, realized variance provides an approximately unbiased estimator of the conditional variance. Moreover, as the sampling frequency increases, the realized variance generally becomes a more precise measure of the underlying volatility, since the estimation error decreases with m. In practice, however, m must not be too large, to prevent intraday returns from becoming correlated due to market microstructure effects [3].

Many assets are not traded continuously, as markets are closed overnight, over weekends, and on holidays. We therefore decompose the daily return into overnight (close-to-open) and intraday (open-to-close) components,

$$Y_t = \log\left(\frac{P_t^{close}}{P_{t-1}^{close}}\right) = \log\left(\frac{P_t^{open}}{P_{t-1}^{close}}\right) + \log\left(\frac{P_t^{close}}{P_t^{open}}\right) = Y_t^{CO} + Y_t^{OC} \tag{2.1.13}$$

where Y_t^{CO} is the overnight (close-to-open) return and Y_t^{OC} is the intraday (open-to-close) return. Here we see the usefulness of the additive structure of log returns. From this it naturally follows that the daily variance can be decomposed into an overnight and an intraday component, if we assume that the two components are uncorrelated:

$$\sigma_t^2 = \mathrm{Var}(Y_t \mid \mathcal{F}_{t-1}) = \mathrm{Var}(Y_t^{CO} + Y_t^{OC} \mid \mathcal{F}_{t-1}) \tag{2.1.14}$$

$$= \mathrm{Var}(Y_t^{CO} \mid \mathcal{F}_{t-1}) + \mathrm{Var}(Y_t^{OC} \mid \mathcal{F}_{t-1}). \tag{2.1.15}$$

While the intraday variance can be estimated from high-frequency data using realized variance $RV_t^{(m)}$, the overnight variance is difficult to estimate reliably from a single observation[2]. One solution is to simply add the squared overnight return to $RV_t^{(m)}$. The variance estimator is then given by

$$\hat{\sigma}_t^2 = RV_t^{(m)} + (Y_t^{CO})^2. \tag{2.1.16}$$

This approach is problematic, as the squared overnight return is a very noisy estimator of the overnight variance [4].

A better approach is to scale the intraday realized variance, which gives the estimator

$$\hat{\sigma}_t^2 = cRV_t^{(m)}, \qquad c = \frac{\frac{1}{T}\sum_{t=1}^{T}(Y_t - \hat{\mu}_t)^2}{\frac{1}{T}\sum_{t=1}^{T} RV_t^{(m)}}, \tag{2.1.17}$$

where $\hat{\mu}_t$ is the sample mean of the daily returns and T is the total number of days in the sample. Here, c measures the historical ratio of total close-to-close variance to intraday variance. Equivalently, if κ denotes the historical ratio of overnight to intraday variance, then $c = 1 + \kappa$. Thus, the missing overnight variance is estimated as a proportion κ of the intraday realized variance, rather than from the noisy overnight return of each individual day. Such an approach is valid under the assumption that intraday and overnight

[2] In case when high-frequency overnight trading data is available, the overnight variance can be approximated in the same way using intranight squared returns, just like the intraday variance [4].

returns are uncorrelated. Weekends and holidays are naturally treated as extended overnight periods [3, 4].

Overall, when high-frequency intraday observations are available, they provide a direct and increasingly precise proxy for the underlying daily variance. In empirical applications, realized variance can therefore be used in place of the unobserved conditional variance, which provides a way to evaluate the performance of volatility models.

2.2 State-space models

A state-space model describes a dynamical system in which an unobserved (latent) stochastic process drives the evolution of an observed process. The central idea is that the system evolves through hidden states, while only noisy measurements of those states are available.

Let $\{X_t\}_{t\geq 0}$ denote the sequence of random variables representing the latent state process, let $\{Y_t\}_{t\geq 1}$ denote the sequence of random variables representing the observed process, and let T denote the time horizon, hence $t \in \{0, 1, \ldots, T\}$[3]. X_t takes values in a measurable state space $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ and Y_t takes values in $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$. Together, $\{(X_t, Y_t)\}_{t\geq 1}$ forms a stochastic process on the joint space $(\mathcal{X} \times \mathcal{Y}, \mathcal{B}(\mathcal{X}) \otimes \mathcal{B}(\mathcal{Y}))$ [5].

The latent process $\{X_t\}_{t\geq 0}$ is modelled as a Markov process, while the observations $\{Y_t\}_{t\geq 1}$ are conditionally independent given the state process. The model is specified by the following densities:

$$X_0 \sim \chi(x_0 \mid \theta), \tag{2.2.1}$$

$$X_t \mid X_{t-1} = x_{t-1} \sim p(x_t \mid x_{t-1}, \theta), \tag{2.2.2}$$

$$Y_t \mid X_t = x_t \sim f(y_t \mid x_t, \theta), \tag{2.2.3}$$

[3]The latent states are indexed from 0 to T, whereas the observed returns are indexed from 1 to T. This emphasizes that the initial state X_0 is the prior and is not associated with any observation. The indexing could be defined differently (e.g., with observations also starting at $t = 0$), but we adopt this convention throughout.

where χ is the *initial density*, p is the *transition density*, and f is the *observation* or *emission density*. The vector $\theta \in \Theta$ denotes the collection of model parameters [5, 6, 7].

A potential problem with this definition is that conditional distribution of $X_t \mid X_{t-1}$ may not always admit a density. A proper way would be to define the state-space model in a more measure focussed way using probability kernels [5]. However, we will not deal with this more general definition in this thesis, as our models will admit nice densities.

Sometimes, the state-space model is defined in even more general terms, where the latent process and observation process are also dependent on the control input U_t, which constitutes another sequence of random variables [8]. Such definition is more used in control theory and engineering. The control inputs can be though of as external factors that influence the system dynamics and observations. In this thesis, we will not consider control inputs, as we are primarily interested in modelling the latent volatility process and its relationship with observed returns.

The main goal of state-space models is to estimate the latent states based on the observed data. With $X_{i:j} = \{X_i, X_{i+1}, \ldots, X_j\}$ we denote the sequence of random variables from time i to j, and with $x_{i:j} = \{x_i, x_{i+1}, \ldots, x_j\}$ we denote their realizations. From the joint posterior distribution of the latent states and model parameters given the observed data, $p(x_{0:t}, \theta \mid y_{1:t})$, we can compute the following marginal distributions of interest:

- the *smoothing distribution*

$$p(x_{0:t} \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}\theta, \tag{2.2.4}$$

- the *filtering distribution*

$$p(x_t \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}x_{0:t-1} \, \mathrm{d}\theta, \tag{2.2.5}$$

- the *prediction distribution*

$$p(x_{t+1} \mid y_{1:t}) = \int p(x_{t+1} \mid x_t, \theta) p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}x_{0:t} \, \mathrm{d}\theta, \tag{2.2.6}$$

- the *posterior distribution of the parameters*

$$p(\theta \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) \, \mathrm{d}x_{0:t}. \tag{2.2.7}$$

A special case of state-space models are hidden Markov models (HMMs). Although there is some discrepancy in the literature, the term hidden Markov model often refers to a state-space model in which the latent state is discrete and takes values in a finite set $\{1, 2, \ldots, K\}$, where K is the number of states [5].

2.3 Markov switching stochastic volatility model

The Markov switching stochastic volatility (MSSV) model is the main model of interest in this thesis. It is a state-space model in which the hidden state is the tuple $X_t = (H_t, S_t)$. The log volatility H_t is continuous, while the regime indicator S_t is discrete and takes values in $\{1, 2, \ldots, K\}$, where K is the number of regimes [9]. The latent state space is therefore

$$\mathcal{X} = \mathbb{R} \times \{1, 2, \ldots, K\}, \tag{2.3.1}$$

equipped with the product σ-algebra $\mathcal{B}(\mathbb{R}) \otimes 2^{\{1,2,\ldots,K\}}$.

- **Observation equation:**

  The observation distribution is defined as

$$Y_t \mid X_t = (h_t, s_t) \sim \mathcal{N}\left(0, \exp\left(h_t\right)\right), \tag{2.3.2}$$

  where $\mathcal{N}$ denotes the normal distribution with the given mean and variance [9, 10]. This distribution admits a density function, which we denote by $f(y_t \mid x_t, \theta) = f(y_t \mid h_t)$, as the observation density does not depend on the regime s_t or on any of the model parameters θ.

  Note that mathematically speaking the latent state H_t is the log variance of the return Y_t, but it is often referred to as the log volatility in

the literature [10, 11, 12]. We will also follow this convention and refer to H_t as the log volatility.

- **Transition equation:**

  The transition distribution is defined as

$$H_t \mid H_{t-1} = h_{t-1}, S_t = s_t \sim \mathcal{N}(\mu_{s_t} + \phi(h_{t-1} - \mu_{s_t}), \eta^2), \tag{2.3.3}$$

$$S_t \mid S_{t-1} = s_{t-1} \sim \text{Categorical}([p_{s_{t-1},1}, \ldots, p_{s_{t-1},K}]), \tag{2.3.4}$$

  where μ_{s_t} denotes the regime-specific mean associated with state s_t, ϕ is the persistence parameter, and η^2 is the variance of the log-volatility innovation process [9]. The values $p_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i)$ are the probabilities of transitioning from regime i to regime j, and they are collected in the $K \times K$ regime transition matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{bmatrix}, \tag{2.3.5}$$

  with the rows of P summing to one, i.e., $\sum_{j=1}^{K} p_{ij} = 1$ for all i. Observe that the log volatility process is essentially an AR(1) process with regime-specific means.

  At first, it may seem that such a conditional definition breaks the Markov property of the latent state process, as H_t depends on S_t. However, the latent state process is still Markovian, as the transition distribution of the joint state (H_t, S_t) depends only on the previous joint state (H_{t-1}, S_{t-1}). This can be seen by factorizing the transition distribution as

$$p(x_t \mid x_{t-1}, \theta) = p((h_t, s_t) \mid (h_{t-1}, s_{t-1}), \theta) \tag{2.3.6}$$

$$= p(s_t \mid h_{t-1}, s_{t-1}, \theta) p(h_t \mid h_{t-1}, s_{t-1}, s_t, \theta) \tag{2.3.7}$$

$$= p(s_t \mid s_{t-1}, \theta) p(h_t \mid h_{t-1}, s_t, \theta). \tag{2.3.8}$$

- **Initial distribution:**

  The MSSV model is completed by specifying the initial distribution as

$$H_0 \mid S_0 = s_0 \sim \mathcal{N}\left(\mu_{s_0}, \frac{\eta^2}{1-\phi^2}\right), \tag{2.3.9}$$

$$S_0 \sim \text{Categorical}\left(\frac{1}{K}, \frac{1}{K}, \ldots, \frac{1}{K}\right), \tag{2.3.10}$$

  that is, according to the stationary distribution of the corresponding AR(1) component, with mean μ_{s_0} and variance $\frac{\eta^2}{1-\phi^2}$, and a uniform prior over the regimes, respectively [9].

  The initial distribution admits a density function, which can be written as $\chi(x_0 \mid \theta) = \chi(s_0 \mid \theta)\chi(h_0 \mid s_0, \theta)$.

In order for the MSSV model to be well-defined, the following assumptions are made:

1. The persistence ϕ lies in the range $(-1, 1)$.
2. Each regime should correspond to at least one time point.
3. The regime means are ordered such that $\mu_1 < \mu_2 < \ldots < \mu_K$.

Assumption 1 ensures that the AR(1) process is stationary. This is needed to ensure that the log volatility does not diverge to infinity. Assumption 2 is necessary to avoid empty regimes, as otherwise some μ_i would be unidentified. Assumption 3 is a common identification constraint in mixture models, which prevents label switching and ensures that the regimes are uniquely defined [9].

In this thesis, we ensure the identifiability of the regimes using the following parameterization:

$$\mu_i = \mu_{i-1} + e^{\delta_i}, \qquad i = 2, \ldots, K, \tag{2.3.11}$$

where $\delta_i \in \mathbb{R}$. This parameterization is motivated by the fact that the exponential function is strictly positive, which guarantees that $\mu_i > \mu_{i-1}$ while leaving δ_i unconstrained.

The parameters of the MSSV model are then given by the vector

$$\theta = \{\mu_1, \delta_2, \ldots, \delta_K, \phi, \eta^2, P\}. \tag{2.3.12}$$

Since P is row stochastic, it has $K(K-1)$ free parameters. Therefore, the total number of free parameters is $K + 2 + K(K-1) = K^2 + 2$.

Note that in the case of $K = 1$, the MSSV model reduces to the standard stochastic volatility (SV) model, in which only the log volatility H_t is a latent variable and there are no regimes.

2.4 Problem formulation

The goal of this thesis is to estimate volatility in financial markets using the MSSV model with parameters θ. Let $y_{1:t} = \{y_1, y_2, \ldots, y_t\}$ denote the sequence of observed returns up to time $t \leq T$. Similarly, let $x_{0:t} = \{x_0, x_1, \ldots, x_t\}$ denote the sequence of latent states up to time t, where x_0 is the initial prior state. In general, we can distinguish between two versions of the problem.

1. **With known parameters**

   If we assume that the model parameters θ are known, then the main objective is to estimate

   $$p(x_{0:t} \mid y_{1:t}, \theta), \tag{2.4.1}$$

   which we can expand using Bayes' theorem:

   $$p(x_{0:t} \mid y_{1:t}, \theta) \propto p(y_{1:t} \mid x_{0:t}, \theta) p(x_{0:t} \mid \theta). \tag{2.4.2}$$

We can further expand the likelihood:

$$p(y_{1:t} \mid x_{0:t}, \theta) \propto f(y_t \mid x_{0:t}, y_{1:t-1}, \theta) p(y_{1:t-1} \mid x_{0:t}, \theta) \tag{2.4.3}$$

$$\propto f(y_t \mid x_t, \theta) f(y_{t-1} \mid x_{0:t}, y_{1:t-2}, \theta) p(y_{1:t-2} \mid x_{0:t}, \theta) \tag{2.4.4}$$

$$\propto \prod_{i=1}^{t} f(y_i \mid x_i, \theta) \tag{2.4.5}$$

$$= \prod_{i=1}^{t} f(y_i \mid x_i), \tag{2.4.6}$$

where we used the fact that each observation y_i depends only on the current latent state x_i (in fact, only on h_i), and not on the entire history of latent states, observations, or parameters. Similarly, we can expand

$$p(x_{0:t} \mid \theta) \propto p(x_t \mid x_{0:t-1}, \theta) p(x_{0:t-1} \mid \theta) \tag{2.4.7}$$

$$\propto p(x_t \mid x_{t-1}, \theta) \cdots p(x_1 \mid x_0, \theta) \chi(x_0 \mid \theta) \tag{2.4.8}$$

$$= \prod_{i=1}^{t} p(x_i \mid x_{i-1}, \theta) \chi(x_0 \mid \theta), \tag{2.4.9}$$

where we used the Markov property of the latent state process.

The final expression is thus

$$p(x_{0:t} \mid y_{1:t}, \theta) \propto \chi(x_0 \mid \theta) \prod_{i=1}^{t} p(x_i \mid x_{i-1}, \theta) f(y_i \mid x_i). \tag{2.4.10}$$

2. **With unknown parameters**

   In reality, the parameters θ of the MSSV model are unknown, so we would also like to estimate them. The main objective is then the joint posterior distribution of the latent states and model parameters given the observed data,

   $$p(x_{0:t}, \theta \mid y_{1:t}), \tag{2.4.11}$$

   from which we can compute the smoothing, filtering and prediction distributions of the latent states and the posterior distribution of the parameters, as described in Section 2.2.

Expanding the joint posterior distribution using Bayes' theorem gives

$$p(x_{0:t}, \theta \mid y_{1:t}) = \frac{p(x_{0:t}, y_{1:t} \mid \theta)p(\theta)}{p(y_{1:t})} \tag{2.4.12}$$

$$\propto p(y_{1:t} \mid x_{0:t}, \theta)p(x_{0:t} \mid \theta)p(\theta). \tag{2.4.13}$$

Reusing the previously derived expressions, the joint posterior distribution can then be expressed as

$$p(x_{0:t}, \theta \mid y_{1:t}) \propto p(\theta)\chi(x_0 \mid \theta) \prod_{i=1}^{t} p(x_i \mid x_{i-1}, \theta) f(y_i \mid x_i). \tag{2.4.14}$$

Equations (2.4.10) and (2.4.14) both reveal the sequential nature of the problem. If we initially sample the prior latent state x_0 (and the model parameters θ, if they are unknown), we can sequentially propagate the latent states through the transition density and update them with new observations through the observation density. This is the main idea behind sequential Monte Carlo methods, which are presented in the next chapter.

2.4.1 Prior distributions of the parameters

In the case of unknown model parameters θ, we need to specify their prior distributions. We assume that the priors of the parameters are independent, so that the joint prior distribution can be factorized as

$$p(\theta) = p(\mu_1) \prod_{i=2}^{K} p(\delta_i)p(\phi)p(\eta^2)p(P). \tag{2.4.15}$$

Even though the model is defined in terms of the parameters δ_i and ϕ, we define the priors in terms of $\Delta_i = \mu_i - \mu_{i-1} = e^{\delta_i}$ and $\varphi = \frac{\phi+1}{2}$. The reason for reparameterizing ϕ as φ is that φ lies in the range $(0, 1)$, which allows us to use the Beta distribution as a prior. The reason for reparameterizing δ_i as Δ_i is that we can then use a truncated normal distribution as a prior, which is more convenient for conditional sampling later on.

We choose the following prior distributions for the model parameters:

$$\mu_1 \sim \mathcal{N}(a_0, 5^2), \tag{2.4.16}$$

$$\Delta_i \sim \mathcal{TN}_{(0,\infty)}(0, 5^2), \qquad i = 2, \ldots, K, \tag{2.4.17}$$

$$\varphi \sim \mathcal{B}(6, 2), \tag{2.4.18}$$

$$\eta^2 \sim \mathcal{IG}(2.001, 0.1), \tag{2.4.19}$$

$$P_{i,:} \sim \mathcal{D}(\alpha_{i,1}, \ldots, \alpha_{i,K}), \qquad i = 1, \ldots, K, \tag{2.4.20}$$

where $\alpha_{i,j} = 1.5$ for $i \neq j$ and $\alpha_{i,i} = 40$. Here, $\mathcal{TN}_{(0,\infty)}$ denotes the truncated normal distribution with support $(0, \infty)$, $\mathcal{B}$ denotes the Beta distribution, $\mathcal{IG}$ denotes the inverse gamma distribution, and $\mathcal{D}$ denotes the Dirichlet distribution.

The priors are chosen to be informative while allowing sufficient flexibility. For μ_1, a_0 is determined through prior elicitation, while the variance 5^2 allows for substantial uncertainty. The prior for Δ_i allows for a wide range of possible values, but still encourages the regime means not to be too far apart. The $\mathcal{B}(6, 2)$ prior for φ favors positive persistence, as is common in financial econometrics. Additionally, previous results from [13] suggest that the choice of prior for this parameter has little effect on the posterior, so we chose hyperparameters that lie in between the values considered there. For η^2, we use $\mathcal{IG}(2.001, 0.1)$, following a combination of the values considered in [10, 13]. This prior has a large variance, allowing for a wide range of values. For the transition matrix P, we depart from the commonly used symmetric Dirichlet prior with hyperparameters 0.5 [9, 10, 13]. Such a prior places substantial mass near 0 and 1, which may encourage regimes to either disappear or become excessively persistent. Instead, motivated by [14], we use $\alpha_{i,i} = 40$ and $\alpha_{i,j} = 1.5$ for $i \neq j$. In the two-regime case, this corresponds to a $\mathcal{B}(40, 1.5)$ prior for the diagonal transition probabilities, with mean 0.964, implying an expected regime duration of approximately 28 days. We find this expected duration reasonable for financial data.

Chapter 3

Sequential Monte Carlo Methods

Sequential Monte Carlo (SMC) methods provide a flexible framework for approximating probability distributions that evolve sequentially over time. They are particularly useful for inference in state-space models, where the latent states are typically not directly observed and their posterior distributions are analytically intractable. SMC methods approximate these distributions using a set of weighted samples, commonly referred to as particles, and update this approximation as new observations become available.

This chapter first introduces the foundations of SMC, beginning with Monte Carlo methods, importance sampling and sequential importance sampling. The problem of weight degeneracy is then addressed through resampling, leading to sequential importance sampling with resampling, also known as particle filtering. The bootstrap particle filter is presented in more detail.

Next, the chapter considers particle-based methods for Bayesian parameter inference. Particle marginal Metropolis-Hastings and particle Gibbs are introduced as particle Markov chain Monte Carlo methods. Finally, SMC^2 is presented as an extension that combines particle filtering with sequential inference for static parameters and latent states.

3.1 Monte Carlo methods

Monte Carlo methods approximate quantities involving a probability distribution by using a finite collection of random samples. The basic idea is simple: rather than evaluating an integral directly, we replace it by an average over samples drawn from the distribution of interest.

Until further notice, let X be a random variable with probability density function $p(x)$, and let g be a measurable function such that $E_p[|g(X)|] < \infty$. The quantity of interest is

$$I = E_p[g(X)] = \int g(x)p(x)\,\mathrm{d}x. \tag{3.1.1}$$

Definition 3.1.1. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} p(x)$. The *Monte Carlo estimator* of I is defined as

$$\hat{I}_N = \frac{1}{N}\sum_{i=1}^{N} g(X_i). \tag{3.1.2}$$

Such an estimator has several desirable properties, including unbiasedness, consistency, and asymptotic normality.

Proposition 3.1.1. *The Monte Carlo estimator $\hat{I}_N$ is unbiased.*

Proof. The proof follows from the linearity of expectation and the fact that the samples are identically distributed.

$$E_p[\hat{I}_N] = E_p\left[\frac{1}{N}\sum_{i=1}^{N} g(X_i)\right] \tag{3.1.3}$$

$$= \frac{1}{N}\sum_{i=1}^{N} E_p[g(X_i)] \tag{3.1.4}$$

$$= \frac{1}{N}\sum_{i=1}^{N} I = I. \tag{3.1.5}$$

□

Proposition 3.1.2. *The Monte Carlo estimator $\hat{I}_N$ is consistent.*

Proof. Because the samples are i.i.d. and $E_p[|g(X)|] < \infty$, the strong law of large numbers applies, giving

$$\hat{I}_N \xrightarrow[N\to\infty]{a.s.} I. \tag{3.1.6}$$

Almost sure convergence implies convergence in probability, so the estimator is consistent. □

Proposition 3.1.3. *If $g(X)$ has finite variance, then the Monte Carlo estimator $\hat{I}_N$ is asymptotically normal.*

Proof. If $g(X)$ has finite variance σ^2, then the central limit theorem applies, giving

$$\sqrt{N}\left(\hat{I}_N - I\right) \xrightarrow[N\to\infty]{D} \mathcal{N}(0, \sigma^2), \tag{3.1.7}$$

which is the definition of asymptotic normality. □

3.1.1 Empirical distribution

The Monte Carlo estimator can be interpreted as integration with respect to an empirical approximation constructed from the samples.

Definition 3.1.2. Given samples $X_1, \ldots, X_N \overset{\text{iid}}{\sim} p(x)$, the *empirical distribution* associated with these samples is defined as

$$\hat{p}_N(x) = \frac{1}{N}\sum_{i=1}^{N} \delta(x - X_i), \tag{3.1.8}$$

where $\delta(\cdot)$ denotes the Dirac delta function.

The empirical distribution assigns equal weight $1/N$ to each sampled value. It therefore provides a discrete approximation to the distribution represented by $p(x)$. The expression in (3.1.8) should be understood through integration rather than as an ordinary probability density with respect to Lebesgue measure.

Proposition 3.1.4. *For g, $\hat{I}_N$ and $\hat{p}_N$ as previously defined,*

$$\hat{I}_N = \int g(x)\hat{p}_N(x)\,\mathrm{d}x. \tag{3.1.9}$$

Proof.

$$\int g(x)\hat{p}_N(x)\,\mathrm{d}x = \int g(x)\frac{1}{N}\sum_{i=1}^{N}\delta(x - X_i)\,\mathrm{d}x \tag{3.1.10}$$

$$= \frac{1}{N}\sum_{i=1}^{N}\int g(x)\delta(x - X_i)\,\mathrm{d}x \tag{3.1.11}$$

$$= \frac{1}{N}\sum_{i=1}^{N} g(X_i) = \hat{I}_N, \tag{3.1.12}$$

where the defining property of the Dirac delta function has been used. □

It is important to distinguish the empirical distribution $\hat{p}_N$ from the density p. Even when p is continuous, $\hat{p}_N$ consists of point masses for every finite N and is therefore not a pointwise approximation of $p(x)$. The approximation is instead meaningful through quantities such as integrals and expectations.

3.2 Importance sampling

The standard Monte Carlo method requires samples from the target distribution p. In many problems, however, sampling directly from p is difficult or even impossible. Importance sampling addresses this problem by introducing a second distribution from which sampling is easier.

Definition 3.2.1. Let $q(x)$ be a proposal distribution satisfying $q(x) > 0$ whenever $p(x) > 0$. The *importance weight* associated with a point x is defined as

$$w(x) = \frac{p(x)}{q(x)}. \tag{3.2.1}$$

The expectation of g under p can then be rewritten as

$$I = E_p[g(X)] = \int g(x)p(x)\,\mathrm{d}x \tag{3.2.2}$$

$$= \int g(x)\frac{p(x)}{q(x)}q(x)\,\mathrm{d}x \tag{3.2.3}$$

$$= E_q\left[g(X)w(X)\right]. \tag{3.2.4}$$

Thus, although samples are generated from q, the importance weights correct for the difference between q and p.

Definition 3.2.2. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $w(X_1), \ldots, w(X_N)$ their corresponding importance weights. The *importance sampling estimator* of $I = E_p[g(X)]$ is defined by

$$\hat{I}_N^{\text{IS}} = \frac{1}{N} \sum_{i=1}^{N} g(X_i) w(X_i). \tag{3.2.5}$$

Proposition 3.2.1. *The importance sampling estimator $\hat{I}_N^{\text{IS}}$ is unbiased.*

Proof.

$$E_q[\hat{I}_N^{\text{IS}}] = \frac{1}{N} \sum_{i=1}^{N} E_q[g(X_i) w(X_i)] \tag{3.2.6}$$

$$= E_q \left[g(X) \frac{p(X)}{q(X)} \right] \tag{3.2.7}$$

$$= \int g(x) \frac{p(x)}{q(x)} q(x) \, \mathrm{d}x = I. \tag{3.2.8}$$

□

Proposition 3.2.2. *The importance sampling estimator $\hat{I}_N^{\text{IS}}$ is consistent.*

Proof. Because $E_q[|g(X)w(X)|] = E_p[|g(X)|] < \infty$, the strong law of large numbers gives

$$\hat{I}_N^{\text{IS}} \xrightarrow[N \to \infty]{a.s.} I, \tag{3.2.9}$$

which means that the estimator is consistent. □

Although the importance sampling estimator is unbiased and consistent for any proposal distribution satisfying the support condition, its variance depends strongly on the choice of q.

3.2.1 (Weighted) empirical distribution

There are two equivalent ways to interpret the importance sampling estimator. The first one follows the same logic as in Section 3.1.1:

$$\hat{I}_N^{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N} w(X_i)g(X_i) = \frac{1}{N}\sum_{i=1}^{N}\int w(x)g(x)\delta(x - X_i)\,\mathrm{d}x \tag{3.2.10}$$

$$= \int w(x)g(x)\frac{1}{N}\sum_{i=1}^{N}\delta(x - X_i)\,\mathrm{d}x \tag{3.2.11}$$

$$= \int w(x)g(x)\hat{q}_N(x)\,\mathrm{d}x, \tag{3.2.12}$$

where $\hat{q}_N$ is the empirical distribution of the samples $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$, so the estimator is simply the expectation of $w(X)g(X)$ with respect to $\hat{q}_N$.

There is another interpretation that is particularly useful. Instead of viewing the weights as part of the integrand, we can absorb them into the empirical distribution itself.

$$\hat{I}_N^{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N} w(X_i)g(X_i) = \frac{1}{N}\sum_{i=1}^{N} w(X_i)\int g(x)\delta(x - X_i)\,\mathrm{d}x \tag{3.2.13}$$

$$= \int g(x)\frac{1}{N}\sum_{i=1}^{N} w(X_i)\delta(x - X_i)\,\mathrm{d}x \tag{3.2.14}$$

$$= \int g(x)\hat{p}_N^{\mathrm{IS}}(x)\,\mathrm{d}x, \tag{3.2.15}$$

leading us to next definition.

Definition 3.2.3. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $w(X_1), \ldots, w(X_N)$ their corresponding importance weights. The *weighted empirical distribution* is defined as

$$\hat{p}_N^{\mathrm{IS}}(x) = \frac{1}{N}\sum_{i=1}^{N} w(X_i)\delta(x - X_i). \tag{3.2.16}$$

This provides an alternative interpretation of importance sampling: although the samples are drawn from the proposal distribution q, the importance weights allow the samples to represent the target distribution p. Unlike

ordinary Monte Carlo, where every sample contributes the same mass $1/N$, importance sampling assigns the sample X_i a mass proportional to $w(X_i)$.

The weighted empirical distribution in (3.2.16), however, is not generally a probability distribution for a finite sample. Its total mass is

$$\int \hat{p}_N^{\mathrm{IS}}(x)\,\mathrm{d}x = \frac{1}{N}\sum_{i=1}^{N} w(X_i), \tag{3.2.17}$$

which is a random quantity and is not necessarily equal to one. Nevertheless, the total mass has expectation equal to one. Since $X \sim q$,

$$E_q[w(X)] = \int \frac{p(x)}{q(x)} q(x)\,\mathrm{d}x = \int p(x)\,\mathrm{d}x = 1, \tag{3.2.18}$$

$$E_q\left[\frac{1}{N}\sum_{i=1}^{N} w(X_i)\right] = \frac{1}{N}\sum_{i=1}^{N} E_q[w(X_i)] = 1. \tag{3.2.19}$$

Additionally, the strong law of large numbers gives

$$\frac{1}{N}\sum_{i=1}^{N} w(X_i) \xrightarrow[N\to\infty]{a.s.} E_q[w(X)] = 1. \tag{3.2.20}$$

Thus, although the weighted empirical distribution does not necessarily have unit mass for a finite sample, its total mass converges almost surely to one as the number of samples increases.

3.3 Self-normalized importance sampling

The previous construction suggests a natural modification: instead of allowing the weights to have a random total mass, normalize them. Writing $w_i = w(X_i)$ for brevity, the *normalized importance weights* are

$$\tilde{w}_i = \frac{w_i}{\sum_{j=1}^{N} w_j}, \tag{3.3.1}$$

which leads to the following definition.

Definition 3.3.1. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $\tilde{w}_1, \ldots, \tilde{w}_N$ their corresponding normalized importance weights. The *self-normalized weighted empirical distribution* is defined as

$$\hat{p}_N^{\text{SNIS}}(x) = \sum_{i=1}^{N} \tilde{w}_i \delta(x - X_i). \tag{3.3.2}$$

This distribution clearly has a total mass of one.

Definition 3.3.2. Let $X_1, \ldots, X_N \overset{\text{iid}}{\sim} q$ and $w_1, \ldots, w_N$ their corresponding importance weights. The *self-normalized importance sampling estimator* of $I = E_p[g(X)]$ is defined as

$$\hat{I}_N^{\text{SNIS}} = \frac{\sum_{i=1}^{N} g(X_i) w_i}{\sum_{i=1}^{N} w_i}. \tag{3.3.3}$$

Proposition 3.3.1. *For g, $\hat{I}_N^{\text{SNIS}}$ and $\hat{p}_N^{\text{SNIS}}$ as previously defined,*

$$\hat{I}_N^{\text{SNIS}} = \int g(x) \hat{p}_N^{\text{SNIS}}(x) \, \mathrm{d}x. \tag{3.3.4}$$

Proof.

$$\int g(x) \hat{p}_N^{\text{SNIS}}(x) \, \mathrm{d}x = \int g(x) \sum_{i=1}^{N} \tilde{w}_i \delta(x - X_i) \, \mathrm{d}x \tag{3.3.5}$$

$$= \sum_{i=1}^{N} \tilde{w}_i \int g(x) \delta(x - X_i) \, \mathrm{d}x \tag{3.3.6}$$

$$= \sum_{i=1}^{N} \tilde{w}_i g(X_i) \tag{3.3.7}$$

$$= \frac{\sum_{i=1}^{N} g(X_i) w_i}{\sum_{i=1}^{N} w_i}. \tag{3.3.8}$$

□

There is a fundamental difference between $\hat{I}_N^{\text{IS}}$ and $\hat{I}_N^{\text{SNIS}}$. The ordinary importance sampling estimator is unbiased, whereas the self-normalized estimator is generally biased for finite N.

Proposition 3.3.2. *The self-normalized importance sampling estimator is, in general, biased.*

Proof. In general, the expected value of a ratio is not equal to the ratio of expected values, so

$$E_q\left[\frac{\sum_{i=1}^N g(X_i)w_i}{\sum_{i=1}^N w_i}\right] \neq \frac{E_q\left[\frac{1}{N}\sum_{i=1}^N g(X_i)w_i\right]}{E_q\left[\frac{1}{N}\sum_{i=1}^N w_i\right]}, \tag{3.3.9}$$

$$E_q\left[\hat{I}_N^{\text{SNIS}}\right] \neq \frac{E_q\left[\hat{I}_N^{\text{IS}}\right]}{1} = I. \tag{3.3.10}$$

□

Proposition 3.3.3. *The self-normalized importance sampling estimator is consistent.*

Proof. The strong law of large numbers gives

$$\frac{1}{N}\sum_{i=1}^N g(X_i)w_i \xrightarrow[N\to\infty]{a.s.} I, \tag{3.3.11}$$

$$\frac{1}{N}\sum_{i=1}^N w_i \xrightarrow[N\to\infty]{a.s.} 1. \tag{3.3.12}$$

Consequently,

$$\hat{I}_N^{\text{SNIS}} = \frac{\frac{1}{N}\sum_{i=1}^N g(X_i)w_i}{\frac{1}{N}\sum_{i=1}^N w_i} \xrightarrow[N\to\infty]{a.s.} I. \tag{3.3.13}$$

□

At first, it may seem pointless to use the self-normalized estimator merely to ensure that $\hat{p}_N^{\text{SNIS}}$ is a proper probability distribution, since the estimator loses its unbiasedness. However, in many applications, especially in Bayesian inference, the target density $p(x)$ is known only up to a proportionality constant. Suppose that the target has the form

$$p(x) = \frac{\tilde{p}(x)}{Z}, \tag{3.3.14}$$

where $\tilde{p}(x)$ is known and Z is an intractable normalization constant. The importance weights are then

$$w(x) = \frac{p(x)}{q(x)} = \frac{\tilde{p}(x)}{Zq(x)}, \tag{3.3.15}$$

which makes it impossible to compute the ordinary importance sampling estimator $\hat{I}_N^{\text{IS}}$. However, the normalized weights are still computable:

$$\tilde{w}_i = \frac{w_i}{\sum_{j=1}^N w_j} = \frac{\frac{\tilde{p}(X_i)}{Zq(X_i)}}{\sum_{j=1}^N \frac{\tilde{p}(X_j)}{Zq(X_j)}} = \frac{\frac{\tilde{p}(X_i)}{q(X_i)}}{\sum_{j=1}^N \frac{\tilde{p}(X_j)}{q(X_j)}}. \tag{3.3.16}$$

Therefore, the self-normalized importance sampling estimator $\hat{I}_N^{\text{SNIS}}$ can be used even when the target density is known only up to a constant of proportionality, at the cost of introducing bias.

3.4 Sequential importance sampling

Sequential importance sampling (SIS) applies importance sampling to a sequence of random variables by exploiting the sequential structure of the problem. Because the main focus of this thesis is on state-space models, in particular the MSSV model, we consider sequential importance sampling in the context of state-space models.

For the moment, assume that the model parameters θ are known. Given the observations $y_{1:T}$, the objective is to approximate the posterior distribution of the latent states, $p(x_{0:T} \mid y_{1:T}, \theta)$. As derived in Section 2.4, the joint density is

$$p(x_{0:T}, y_{1:T} \mid \theta) = \chi(x_0 \mid \theta) \prod_{t=1}^T f(y_t \mid x_t, \theta) p(x_t \mid x_{t-1}, \theta). \tag{3.4.1}$$

Until further notice, assume that the proposal distribution can also be factorized sequentially as

$$q(x_{0:T} \mid y_{1:T}, \theta) = q_0(x_0 \mid \theta) \prod_{t=1}^T q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta), \tag{3.4.2}$$

from which we can sample entire trajectories of the latent states, conditioned on the currently available observations, in a sequential manner.

The importance weight can therefore be computed recursively as

$$w_T = \frac{p(x_{0:T}, y_{1:T} \mid \theta)}{q(x_{0:T} \mid y_{1:T}, \theta)} \tag{3.4.3}$$

$$= \frac{\chi(x_0 \mid \theta) \prod_{t=1}^{T} f(y_t \mid x_t, \theta) p(x_t \mid x_{t-1}, \theta)}{q_0(x_0 \mid \theta) \prod_{t=1}^{T} q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta)} \tag{3.4.4}$$

$$= w_{T-1} \cdot \frac{f(y_T \mid x_T, \theta) p(x_T \mid x_{T-1}, \theta)}{q_T(x_T \mid x_{0:T-1}, y_{1:T}, \theta)} \tag{3.4.5}$$

$$= w_{T-1} \alpha_T = \prod_{t=0}^{T} \alpha_t \tag{3.4.6}$$

with $\alpha_0 = w_0 = \frac{\chi(x_0|\theta)}{q_0(x_0|\theta)}$.

Definition 3.4.1. The *marginal likelihood* is defined as

$$p(y_{1:T} \mid \theta) = \int p(x_{0:T}, y_{1:T} \mid \theta) \, \mathrm{d}x_{0:T}. \tag{3.4.7}$$

Introducing the importance weights, the marginal likelihood can therefore be written as

$$p(y_{1:T} \mid \theta) = \int w_T \, q(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T} = E_q[w_T]. \tag{3.4.8}$$

For nonlinear and non-Gaussian state-space models, the marginal likelihood is typically intractable and therefore has to be estimated.

Definition 3.4.2. Let $X_{0:T}^{(1)}, \ldots, X_{0:T}^{(N)} \overset{\text{iid}}{\sim} q$ be sampled trajectories and $w_T^{(i)}$ their corresponding importance weights. Then the *marginal likelihood estimator* is defined as

$$\hat{p}(y_{1:T} \mid \theta) = \frac{1}{N} \sum_{i=1}^{N} w_T^{(i)}. \tag{3.4.9}$$

As a result the marginal likelihood estimator can therefore be written as

$$\hat{p}(y_{1:T} \mid \theta) = \frac{1}{N} \sum_{i=1}^{N} \prod_{t=0}^{T} \alpha_t^{(i)}. \tag{3.4.10}$$

Proposition 3.4.1. *The marginal likelihood estimator $\hat{p}(y_{1:T} \mid \theta)$ is unbiased and consistent.*

Proof. Unbiasedness follows from

$$E_q[\hat{p}(y_{1:T} \mid \theta)] = E_q\left[\frac{1}{N}\sum_{i=1}^{N} w_T^{(i)}\right] = \frac{1}{N}\sum_{i=1}^{N} E_q[w_T^{(i)}] \tag{3.4.11}$$

$$= E_q[w_T] = \int w_T q(x_{0:T} \mid y_{1:T}, \theta)\,\mathrm{d}x_{0:T} \tag{3.4.12}$$

$$= \int p(x_{0:T}, y_{1:T} \mid \theta)\,\mathrm{d}x_{0:T} = p(y_{1:T} \mid \theta), \tag{3.4.13}$$

where the linearity of expectation and the fact that the trajectories are identically distributed have been used. Since the weights $w_T^{(i)}$ are i.i.d., non-negative and have finite expectation $p(y_{1:T} \mid \theta)$, consistency follows from the strong law of large numbers,

$$\hat{p}(y_{1:T} \mid \theta) \xrightarrow[N\to\infty]{a.s.} p(y_{1:T} \mid \theta). \tag{3.4.14}$$

Alternatively, $\hat{p}(y_{1:T} \mid \theta)$ is the (ordinary) importance sampling estimator $\hat{I}_N^{\mathrm{IS}}$ with $g \equiv 1$, hence the proofs of Propositions 3.2.1 and 3.2.2 carry over directly. □

3.4.1 Smoothing distribution

Definition 3.4.3. The *smoothing distribution* is the posterior distribution of the entire latent trajectory given all available observations,

$$p(x_{0:T} \mid y_{1:T}, \theta) = \frac{p(x_{0:T}, y_{1:T} \mid \theta)}{p(y_{1:T} \mid \theta)}. \tag{3.4.15}$$

The *smoothing expectation* of a function g is defined as

$$I^{\mathrm{smth}} = E\left[g(X_{0:T}) \mid y_{1:T}, \theta\right] = \int g(x_{0:T}) p(x_{0:T} \mid y_{1:T}, \theta)\,\mathrm{d}x_{0:T}. \tag{3.4.16}$$

Rewriting the smoothing expectation in terms of the proposal distribution gives

$$I^{\mathrm{smth}} = \frac{1}{p(y_{1:T} \mid \theta)} \int g(x_{0:T}) \frac{p(x_{0:T}, y_{1:T} \mid \theta)}{q(x_{0:T} \mid y_{1:T}, \theta)} q(x_{0:T} \mid y_{1:T}, \theta)\,\mathrm{d}x_{0:T} \tag{3.4.17}$$

$$= \frac{E_q\left[g(X_{0:T}) w_T\right]}{p(y_{1:T} \mid \theta)} = \frac{E_q\left[g(X_{0:T}) w_T\right]}{E_q\left[w_T\right]}. \tag{3.4.18}$$

The smoothing distribution is thus a target known only up to the normalizing constant $p(y_{1:T} \mid \theta)$, with unnormalized density $p(x_{0:T}, y_{1:T} \mid \theta)$, which is exactly the setting of Section 3.3. Replacing both expectations by Monte Carlo averages over the sampled trajectories motivates the following definitions.

Definition 3.4.4. Let $X_{0:T}^{(i)} \overset{\text{iid}}{\sim} q$ be sampled trajectories, $w_T^{(i)}$ their importance weights, and

$$\tilde{w}_T^{(i)} = \frac{w_T^{(i)}}{\sum_{j=1}^N w_T^{(j)}} \tag{3.4.19}$$

the corresponding normalized weights. The *smoothing estimator* of I^{smth} is defined as

$$\hat{I}_N^{\text{smth}} = \frac{\sum_{i=1}^N g(X_{0:T}^{(i)}) w_T^{(i)}}{\sum_{i=1}^N w_T^{(i)}} = \sum_{i=1}^N \tilde{w}_T^{(i)} g(X_{0:T}^{(i)}), \tag{3.4.20}$$

inducing the *empirical smoothing distribution* as

$$\hat{p}_N^{\text{smth}}(x_{0:T} \mid y_{1:T}, \theta) = \sum_{i=1}^N \tilde{w}_T^{(i)} \delta(x_{0:T} - X_{0:T}^{(i)}). \tag{3.4.21}$$

Proposition 3.4.2. *For g, $\hat{I}_N^{\text{smth}}$ and $\hat{p}_N^{\text{smth}}$ as previously defined,*

$$\hat{I}_N^{\text{smth}} = \int g(x_{0:T}) \hat{p}_N(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T}. \tag{3.4.22}$$

Moreover, the smoothing estimator $\hat{I}_N^{\text{smth}}$ is, in general, biased, but consistent.

Proof. The first claim follows from the defining property of the Dirac delta function,

$$\int g(x_{0:T}) \hat{p}_N(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T} = \sum_{i=1}^N \tilde{w}_T^{(i)} \int g(x_{0:T}) \delta(x_{0:T} - X_{0:T}^{(i)}) \, \mathrm{d}x_{0:T} \tag{3.4.23}$$

$$= \sum_{i=1}^N \tilde{w}_T^{(i)} g(X_{0:T}^{(i)}) = \hat{I}_N^{\text{smth}}. \tag{3.4.24}$$

The estimator $\hat{I}_N^{\text{smth}}$ is the self-normalized importance sampling estimator $\hat{I}_N^{\text{SNIS}}$ with target $p(x_{0:T} \mid y_{1:T}, \theta)$, unnormalized target $p(x_{0:T}, y_{1:T} \mid \theta)$, normalizing constant $Z = p(y_{1:T} \mid \theta)$ and proposal $q(x_{0:T} \mid y_{1:T}, \theta)$. Bias and consistency therefore follow from Propositions 3.3.2 and 3.3.3. □

3.4.2 Filtering distribution

Definition 3.4.5. The *filtering distribution* is the marginal of the smoothing distribution,

$$p(x_T \mid y_{1:T}, \theta) = \int p(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T-1}. \tag{3.4.25}$$

The *filtering expectation* of a function g is defined as

$$I^{\text{filt}} = E\left[g(X_T) \mid y_{1:T}, \theta\right] = \int g(x_T) p(x_T \mid y_{1:T}, \theta) \, \mathrm{d}x_T. \tag{3.4.26}$$

In the empirical setting, the states at previous times are easily integrated out, since $\delta(x_{0:T} - X_{0:T}^{(i)}) = \delta(x_T - X_T^{(i)}) \, \delta(x_{0:T-1} - X_{0:T-1}^{(i)})$,

$$\int \hat{p}_N(x_{0:T} \mid y_{1:T}, \theta) \, \mathrm{d}x_{0:T-1} \tag{3.4.27}$$

$$= \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_T - X_T^{(i)}) \int \delta(x_{0:T-1} - X_{0:T-1}^{(i)}) \, \mathrm{d}x_{0:T-1} \tag{3.4.28}$$

$$= \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_T - X_T^{(i)}), \tag{3.4.29}$$

which leads to the following definition.

Definition 3.4.6. Let $X_{0:T}^{(i)} \overset{\text{iid}}{\sim} q$ and $\tilde{w}_T^{(i)}$ their corresponding normalized weights. The *filtering estimator* of I^{filt} is defined as

$$\hat{I}_N^{\text{filt}} = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_T^{(i)}), \tag{3.4.30}$$

inducing the *empirical filtering distribution* is as

$$\hat{p}_N^{\text{filt}}(x_T \mid y_{1:T}, \theta) = \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_T - X_T^{(i)}). \tag{3.4.31}$$

Proposition 3.4.3. *The filtering estimator $\hat{I}_N^{\text{filt}}$ is, in general, biased, but consistent.*

Proof. Define the function $\bar{g}(x_{0:T}) = g(x_T)$, which depends on the trajectory only through its final state. Its smoothing expectation is

$$E\left[\bar{g}(X_{0:T}) \mid y_{1:T}, \theta\right] = \int g(x_T)\, p(x_{0:T} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T} \tag{3.4.32}$$

$$= \int g(x_T) \left(\int p(x_{0:T} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T-1} \right) \mathrm{d}x_T \tag{3.4.33}$$

$$= \int g(x_T)\, p(x_T \mid y_{1:T}, \theta)\, \mathrm{d}x_T = I^{\mathrm{filt}}, \tag{3.4.34}$$

where the previous states have been integrated out. Likewise, the smoothing estimator of this expectation is

$$\sum_{i=1}^{N} \tilde{w}_T^{(i)} \bar{g}(X_{0:T}^{(i)}) = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_T^{(i)}) = \hat{I}_N^{\mathrm{filt}}. \tag{3.4.35}$$

Thus, $\hat{I}_N^{\mathrm{filt}}$ is the smoothing estimator $\hat{I}_N^{\mathrm{smth}}$ applied to $\bar{g}$, and its bias and consistency follow from Proposition 3.4.2. □

3.4.3 Predictive distribution

Definition 3.4.7. The (one-step) *predictive distribution* is obtained by propagating the filtering distribution through the transition density,

$$p(x_{T+1} \mid y_{1:T}, \theta) = \int p(x_{T+1} \mid x_T, \theta) p(x_T \mid y_{1:T}, \theta)\, \mathrm{d}x_T. \tag{3.4.36}$$

The *predictive expectation* of a function g is defined as

$$I^{\mathrm{pred}} = E\left[g(X_{T+1}) \mid y_{1:T}, \theta\right] = \int g(x_{T+1}) p(x_{T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{T+1}. \tag{3.4.37}$$

Replacing the filtering distribution with its empirical approximation gives

$$\int p(x_{T+1} \mid x_T, \theta) \hat{p}_N(x_T \mid y_{1:T}, \theta)\, \mathrm{d}x_T = \sum_{i=1}^{N} \tilde{w}_T^{(i)} p(x_{T+1} \mid X_T^{(i)}, \theta), \tag{3.4.38}$$

a mixture of transition densities weighted by the normalized weights. A particle approximation of this mixture is obtained by propagating each particle through its transition density, which leads to the following definition.

Definition 3.4.8. Let $X_{0:T}^{(i)}$ and $\tilde{w}_T^{(i)}$ be as in Definition 3.4.4, and let $X_{T+1}^{(i)} \sim p(x_{T+1} \mid X_T^{(i)}, \theta)$ be sampled independently. The *predictive estimator* of I^{pred} is defined as

$$\hat{I}_N^{\text{pred}} = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_{T+1}^{(i)}), \tag{3.4.39}$$

inducing the *empirical predictive distribution* is as

$$\hat{p}_N(x_{T+1} \mid y_{1:T}, \theta) = \sum_{i=1}^{N} \tilde{w}_T^{(i)} \delta(x_{T+1} - X_{T+1}^{(i)}). \tag{3.4.40}$$

Proposition 3.4.4. *The predictive estimator $\hat{I}_N^{\text{pred}}$ is, in general, biased, but consistent.*

Proof. The extended trajectories $X_{0:T+1}^{(i)}$ are independent draws from the extended proposal

$$\bar{q}(x_{0:T+1} \mid y_{1:T}, \theta) = q(x_{0:T} \mid y_{1:T}, \theta)\, p(x_{T+1} \mid x_T, \theta). \tag{3.4.41}$$

Take as the target the extended smoothing distribution

$$p(x_{0:T+1} \mid y_{1:T}, \theta) = \frac{p(x_{0:T}, y_{1:T} \mid \theta)\, p(x_{T+1} \mid x_T, \theta)}{p(y_{1:T} \mid \theta)}, \tag{3.4.42}$$

whose marginal in x_{T+1} is the predictive distribution, so that

$$I^{\text{pred}} = \int g(x_{T+1})\, p(x_{0:T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T+1}. \tag{3.4.43}$$

The transition densities cancel in the corresponding unnormalized importance weight,

$$\bar{w}_{T+1} = \frac{p(x_{0:T}, y_{1:T} \mid \theta)\, p(x_{T+1} \mid x_T, \theta)}{q(x_{0:T} \mid y_{1:T}, \theta)\, p(x_{T+1} \mid x_T, \theta)} = w_T. \tag{3.4.44}$$

Rewriting the predictive expectation in terms of the extended proposal therefore gives

$$I^{\text{pred}} = \frac{1}{p(y_{1:T} \mid \theta)} \int g(x_{T+1})\, \bar{w}_{T+1}\, \bar{q}(x_{0:T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T+1} \tag{3.4.45}$$

$$= \frac{E_{\bar{q}}\left[g(X_{T+1})\, w_T\right]}{p(y_{1:T} \mid \theta)}. \tag{3.4.46}$$

Analogously, the normalizing constant is

$$p(y_{1:T} \mid \theta) = \int \bar{w}_{T+1}\, \bar{q}(x_{0:T+1} \mid y_{1:T}, \theta)\, \mathrm{d}x_{0:T+1} = E_{\bar{q}}\left[w_T\right], \tag{3.4.47}$$

which yields

$$I^{\mathrm{pred}} = \frac{E_{\bar{q}}\left[g(X_{T+1})\, w_T\right]}{E_{\bar{q}}\left[w_T\right]}. \tag{3.4.48}$$

Replacing both expectations by Monte Carlo averages over $X_{0:T+1}^{(i)} \overset{\text{iid}}{\sim} \bar{q}$ gives

$$\frac{\frac{1}{N}\sum_{i=1}^{N} g(X_{T+1}^{(i)})\, w_T^{(i)}}{\frac{1}{N}\sum_{i=1}^{N} w_T^{(i)}} = \sum_{i=1}^{N} \tilde{w}_T^{(i)} g(X_{T+1}^{(i)}) = \hat{I}_N^{\mathrm{pred}}. \tag{3.4.49}$$

Hence $\hat{I}_N^{\mathrm{pred}}$ is exactly the self-normalized importance sampling estimator $\hat{I}_N^{\mathrm{SNIS}}$ of I^{pred} with proposal $\bar{q}$, unnormalized target $p(x_{0:T}, y_{1:T} \mid \theta)\, p(x_{T+1} \mid x_T, \theta)$ and normalizing constant $Z = p(y_{1:T} \mid \theta)$. Bias and consistency follow from Propositions 3.3.2 and 3.3.3. □

Algorithm 3.4.1 Sequential importance sampling

Require: Observations $y_{1:T}$, number of particles N, proposal densities q_0, q_t, and model densities χ, p, f.

for $i = 1, \ldots, N$ **do**
  Sample $X_0^{(i)} \sim q_0(x_0 \mid \theta)$.
  Set $w_0^{(i)} = \dfrac{\chi(X_0^{(i)} \mid \theta)}{q_0(X_0^{(i)} \mid \theta)}$.
end for
Normalize weights to obtain $\tilde{w}_0^{(i)}$.
for $t = 1, \ldots, T$ **do**
  for $i = 1, \ldots, N$ **do**
    Propagate $X_t^{(i)} \sim q_t(\cdot \mid X_{0:t-1}^{(i)}, y_{1:t}, \theta)$.
    Compute $\alpha_t^{(i)} = \dfrac{f(y_t \mid X_t^{(i)}, \theta) p(X_t^{(i)} \mid X_{t-1}^{(i)}, \theta)}{q_t(X_t^{(i)} \mid X_{0:t-1}^{(i)}, y_{1:t}, \theta)}$.
    Update $w_t^{(i)} = w_{t-1}^{(i)} \alpha_t^{(i)}$.
  end for
  Normalize weights to obtain $\tilde{w}_t^{(i)}$.
end for

In summary, the sequential updates of the importance weights together with the propagation of particles allow us to approximate the smoothing, filtering, and predictive distributions, and their expectations, in a sequential manner. The procedure is summarized in Algorithm 3.4.1. Since all of $\hat{I}_N^{\text{smth}}$, $\hat{I}_N^{\text{filt}}$ and $\hat{I}_N^{\text{pred}}$ are self-normalized, they are biased but consistent estimators. Only the marginal likelihood estimator $\hat{p}(y_{1:T} \mid \theta)$ is unbiased.

The main limitation of SIS is *weight degeneracy*. As the number of observations increases, the importance weights become increasingly unequal, because only a small fraction of the particle trajectories are able to explain all of the observations. Eventually, almost all of the normalized weight is concentrated on a few particles, and the remaining particles contribute virtually nothing to the estimates. This motivates the idea of resampling.

3.5 Sequential importance sampling with resampling

Sequential importance sampling with resampling (SISR), also known as particle filtering, is an extension of SIS that incorporates a resampling step to mitigate the problem of weight degeneracy.

Definition 3.5.1. The *effective sample size* at time t is calculated as

$$\text{ESS}_t = \frac{1}{\sum_{i=1}^{N} \left(\tilde{w}_t^{(i)}\right)^2}, \tag{3.5.1}$$

where $\tilde{w}_t^{(i)}$ are the current normalized importance weights.

Resampling can be performed adaptively, only when the effective sample size falls below a chosen threshold. Note that $1 \leq \text{ESS}_t \leq N$, with $\text{ESS}_t = N$ for equal weights and $\text{ESS}_t = 1$ when all the weight is concentrated on a single particle. In this thesis, however, resampling is performed at every time step.

Mathematically, resampling replaces the empirical distribution

$$\hat{p}_N(x_{0:t} \mid y_{1:t}) = \sum_{i=1}^{N} \tilde{w}_t^{(i)} \delta(x_{0:t} - X_{0:t}^{(i)}) \tag{3.5.2}$$

with a new empirical distribution

$$\hat{p}_N^{\text{res}}(x_{0:t} \mid y_{1:t}) = \frac{1}{N} \sum_{i=1}^{N} N^{(i)} \delta(x_{0:t} - X_{0:t}^{(i)}), \tag{3.5.3}$$

where $N^{(i)}$ is the number of offspring of particle $X_{0:t}^{(i)}$ after resampling, such that $\sum_{i=1}^{N} N^{(i)} = N$. Common resampling algorithms include multinomial, stratified, residual, and systematic resampling. The last of these, systematic resampling, is used in this thesis.

The validity of resampling can be established by considering its effect on estimates of expectations with respect to the weighted empirical distribution. As always let g be a bounded measurable function with finite expectation. Before resampling, its expectation under the weighted empirical distribution is

$$\hat{I}_N = \sum_{i=1}^{N} \tilde{w}_t^{(i)} g(X_{0:t}^{(i)}). \tag{3.5.4}$$

After resampling, the corresponding estimate becomes

$$\hat{I}_N^{\text{res}} = \frac{1}{N} \sum_{i=1}^{N} N^{(i)} g(X_{0:t}^{(i)}). \tag{3.5.5}$$

Although $\hat{I}_N$ and $\hat{I}_N^{\text{res}}$ generally differ for a particular resampling outcome, an unbiased resampling scheme satisfies

$$E\left[N^{(i)} \mid \tilde{w}_t^{(1:N)}\right] = N\tilde{w}_t^{(i)}, \qquad i = 1, \ldots, N, \tag{3.5.6}$$

where the expectation is taken with respect to the randomness introduced by resampling, conditional on the particles and their weights. All four schemes mentioned above satisfy this condition.

Proposition 3.5.1. *Let $\hat{I}_N$ and $\hat{I}_N^{\text{res}}$ be the estimators before and after resampling. For an unbiased resampling scheme satisfying* (3.5.6)*, the resampled estimator is conditionally unbiased,*

$$E\left[\hat{I}_N^{\text{res}} \mid \tilde{w}_t^{(1:N)}\right] = \hat{I}_N, \tag{3.5.7}$$

where the expectation is taken with respect to the randomness introduced by resampling.

Proof. Since the particles $X_{0:t}^{(i)}$ are fixed given $\tilde{w}_t^{(1:N)}$, the values $g(X_{0:t}^{(i)})$ can be taken out of the conditional expectation, giving

$$E\left[\hat{I}_N^{\text{res}} \mid \tilde{w}_t^{(1:N)}\right] = E\left[\frac{1}{N}\sum_{i=1}^{N} N^{(i)} g(X_{0:t}^{(i)}) \,\middle|\, \tilde{w}_t^{(1:N)}\right] \tag{3.5.8}$$

$$= \frac{1}{N}\sum_{i=1}^{N} g(X_{0:t}^{(i)}) E\left[N^{(i)} \mid \tilde{w}_t^{(1:N)}\right] \tag{3.5.9}$$

$$= \frac{1}{N}\sum_{i=1}^{N} g(X_{0:t}^{(i)}) N\tilde{w}_t^{(i)} = \sum_{i=1}^{N} g(X_{0:t}^{(i)})\tilde{w}_t^{(i)} = \hat{I}_N. \tag{3.5.10}$$

□

Resampling creates a new set of equally weighted particles. A slight complication arises when keeping track of the entire trajectories of the latent states, as resampling mixes the trajectories of the particles. This can be solved by keeping track of the ancestry of the particles, which is done by storing the indices of the parent particles during resampling. This allows us to reconstruct the entire trajectory of each particle at any time step. Figure 3.5.1 illustrates the effect of resampling on the particle trajectories. The resampling step prunes particles with low weights and replicates particles with high weights, so that computational effort is concentrated on regions of high posterior probability. A side effect of repeated resampling is *path degeneracy*: going backwards in time, the reconstructed trajectories coalesce to only a few common ancestors, so the approximation of the smoothing distribution $p(x_{0:t} \mid y_{1:t})$ deteriorates for early states, even though the approximation of the filtering distribution remains accurate.

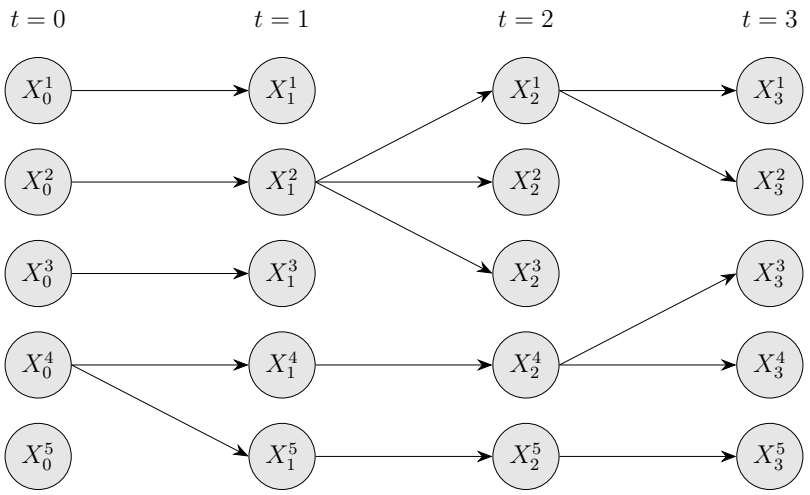


Figure 3.5.1: Schematic illustration of the resampling process. Particles with low weights are pruned, while particles with high weights are replicated.

3.5.1 Bootstrap particle filter

One of the simplest and most widely used particle filtering algorithms is the bootstrap particle filter, which specifies the proposal distribution as

$$q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta) = p(x_t \mid x_{t-1}, \theta), \qquad q_0(x_0 \mid \theta) = \chi(x_0 \mid \theta), \tag{3.5.11}$$

where $p(x_t \mid x_{t-1}, \theta)$ and $\chi(x_0 \mid \theta)$ are the transition and initial densities of the state-space model. The weight update then simplifies to

$$w_t = w_{t-1} \frac{f(y_t \mid x_t, \theta) p(x_t \mid x_{t-1}, \theta)}{q_t(x_t \mid x_{0:t-1}, y_{1:t}, \theta)} = w_{t-1} f(y_t \mid x_t, \theta). \tag{3.5.12}$$

The entire bootstrap particle filter algorithm is presented in Algorithm 3.5.1. Due to the resampling step, the particles at the end of each iteration become equally weighted. Expectations with respect to the empirical distribution can be calculated either by using the weighted particles before resampling or by using the equally weighted particles after resampling. The latter introduces additional Monte Carlo variability due to the randomness of the resampling step, so the former is generally preferred.

Algorithm 3.5.1 Bootstrap particle filter

Require: Observations $y_{1:T}$, number of particles N, model densities χ, p, f.

1: **for** $i = 1, \ldots, N$ **do**
2: Sample $X_0^{(i)} \sim \chi(x_0 \mid \theta)$.
3: Set $w_0^{(i)} = \tilde{w}_0^{(i)} = 1/N$.
4: **end for**
5: **for** $t = 1, \ldots, T$ **do**
6: **for** $i = 1, \ldots, N$ **do**
7: Propagate $X_t^{(i)} \sim p(\cdot \mid X_{t-1}^{(i)}, \theta)$.
8: Compute $\alpha_t^{(i)} = f(y_t \mid X_t^{(i)}, \theta)$.
9: Update $w_t^{(i)} = w_{t-1}^{(i)} \alpha_t^{(i)}$.
10: **end for**
11: **for** $i = 1, \ldots, N$ **do**
12: Normalize weights $\tilde{w}_t^{(i)} = \dfrac{w_t^{(i)}}{\sum_{j=1}^{N} w_t^{(j)}}$.
13: **end for**
14: Resample particles $\{X_t^{(i)}\}_{i=1}^N$ according to $\{\tilde{w}_t^{(i)}\}_{i=1}^N$.
15: Set $w_t^{(i)} = 1/N$ for all $i = 1, \ldots, N$.
16: **end for**

3.6 Particle Markov chain Monte Carlo

3.6.1 Particle marginal Metropolis-Hastings

3.6.2 Particle Gibbs

3.7 SMC^2

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion

Appendix A

Title of the appendix 1

Example of the appendix.

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